

The `what()` Function in C++

The `what()` function is a virtual function defined in the `std::exception` class, which is the base class for all exception classes in C++. It returns a string that describes the error or exceptional event that occurred.

Using the `what()` Function

The `what()` function is used to get a description of an error or exceptional event that occurred during the execution of a program. It is commonly used in exception handling to provide more information about the error or exceptional event that occurred.

Using the `what()` function has several advantages over simply printing a string:

1. It allows you to centralize the error handling code in the catch block, rather than having to duplicate the error message in multiple places in your code.
2. It allows you to customize the error message for different exception types. For example, you might have different exception classes for different types of errors, and each class could have its own implementation of the `what()` function that returns a different error message.
3. It allows you to use the `std::exception` class or one of its derived classes as a catch type, rather than having to catch each exception class individually. This can make your code more flexible and easier to maintain.

```

#include <iostream>
#include <exception>

class DivisionByZeroException : public std::exception {
public:
    virtual const char* what() const throw() {
        return "Division by zero";
    }
};

double divide(double a, double b) {
    if (b == 0) {
        throw DivisionByZeroException();
    }
    return a / b;
}

int main() {
    try {
        std::cout << divide(5, 0) << std::endl;
    } catch (DivisionByZeroException& e) {
        std::cerr << "Error: " << e.what() << std::endl;
    }

    return 0;
}

```

In this example, the `divide()` function throws a `DivisionByZeroException` if the divisor is zero. The exception is caught in the `catch` block, which prints an error message to the standard error stream using the `what()` function of the `DivisionByZeroException` class.